Zero forcing on temporal graphs: Layer-local and footprint-constrained dynamics

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Abstract

Zero forcing is a graph-coloring process in which a blue vertex with a unique white neighbor forces that neighbor to blue. The standard process assumes a fixed graph, whereas many communication, interaction, and control networks change over time. We study two synchronous finite-lifetime extensions to a temporal graph. In *layer-local temporal zero forcing*, uniqueness is evaluated in the current layer. In *footprint-constrained temporal zero forcing*, uniqueness is evaluated in the aggregate footprint while the forcing edge must be currently active. The distinction changes the process fundamentally: layer-local forces may branch over time, while footprint-constrained forces form vertex-disjoint chains.

We prove sharp universal growth bounds, dominance and reversal results, and exact formulas for alternating temporal paths and even cycles. These formulas give linear separations between the two temporal parameters and ordinary zero forcing of the aggregate graph. A temporal-tree family shows that static aggregation can overestimate layer-local forcing by a linear factor and that reversing the layers can change the layer-local number by a factor linear in the order; the footprint-constrained number is invariant under reversal. We also prove that layer-local temporal zero forcing is NP-complete and W[2]-hard by seed budget even when every layer is a star, every edge appears once, and the footprint is bipartite. Finally, the growth bounds yield fixed-parameter algorithms parameterized by seed budget and lifetime. Exhaustive computations verify all closed forms on small instances.

1 Introduction

Zero forcing was introduced as a combinatorial tool for minimum-rank problems and has subsequently been connected with graph searching, propagation, and network controllability [1, 19, 8, 14]. Starting from an initial set of blue vertices, the color-change rule repeatedly allows a blue vertex with exactly one white neighbor to color that neighbor blue. The minimum size of an initial set that eventually colors the graph is the zero-forcing number.

The static graph assumption is restrictive when edges represent contacts or communication opportunities. Temporal-graph models retain the order in which edges are available, and temporal order can change reachability, influence, and controllability even when the aggregate graph is unchanged [4]. Existing work has temporalized target-set selection [17], compared temporal and aggregate influence [13], minimized temporal source sets [18], and studied controllability of time-varying networks [16, 15, 20]. Applying the unique-white-neighbor rule to temporal layers introduces an additional modeling choice: should uniqueness be evaluated among currently active neighbors or among all neighbors that can ever appear?

We formalize both choices. The layer-local rule evaluates uniqueness in the current layer. It models a source whose ambiguity is limited to its current contacts. The footprint-constrained rule evaluates uniqueness in the aggregate footprint and then requires the unique remaining edge to be active. It models a temporally scheduled version of a valid static zero-forcing deduction. The two rules agree on neither their growth mechanism nor their response to aggregation and time reversal.

Our contributions are as follows.

1. We prove

$$Z_{\mathrm{LL}}(\mathcal{G}) \le Z_{\mathrm{FC}}(\mathcal{G}), \qquad Z_{\mathrm{LL}}(\mathcal{G}) \ge \left\lceil \frac{n}{2^T} \right\rceil, \qquad Z_{\mathrm{FC}}(\mathcal{G}) \ge \max\left\{ Z(H), \left\lceil \frac{n}{T+1} \right\rceil \right\},$$

   where T is the lifetime and H is the footprint. Both growth bounds are sharp.

2. For the two-layer alternating path $\mathcal{P}_n$ and even cycle $\mathcal{C}_n$, we determine both parameters exactly:

$$Z_{\mathrm{LL}}(\mathcal{P}_n) = \left\lfloor \frac{n}{4} \right\rfloor + 1, \quad Z_{\mathrm{FC}}(\mathcal{P}_n) = \left\lceil \frac{n}{3} \right\rceil,$$

   and

$$Z_{\mathrm{LL}}(\mathcal{C}_n) = \left\lceil \frac{n}{4} \right\rceil, \quad Z_{\mathrm{FC}}(\mathcal{C}_n) = 2\left\lceil \frac{n}{6} \right\rceil.$$

3. We construct a temporal tree on $n = 2^T$ vertices for which the forward layer-local number is 1, the footprint-constrained number is $n/2$, and the layer-local number after reversing time is $n/2$. Meanwhile, the static zero-forcing number of the footprint is $n/4$. Thus the family gives linear additive aggregation errors in both directions. Together with the alternating paths, it shows that either multiplicative error can grow linearly with n.

4. We prove footprint-constrained forcing is invariant under reversing the layer order, in contrast to the unbounded directional asymmetry of the layer-local rule.

5. We establish restricted NP-completeness and W[2]-hardness for the layer-local problem through an additive optimum-preserving reduction from Set Cover, and derive fixed-parameter algorithms from the two growth laws.

2 Related work and prior-art boundary

Standard zero forcing and its minimum-rank origin are developed in [1]. Parallel rounds and propagation time on a fixed graph are studied in [12], and recent work has introduced relaxed chronologies and parallel increasing path covers [11]. We therefore avoid using "chronological zero forcing" as the name of the new parameters: chronology is already established terminology in the static literature.

A closely related temporal diffusion model is temporal target-set selection. Its update is centered on the target: a vertex activates when sufficiently many of its current neighbors are active [17]. Our condition is centered on the source: a blue source can act only when it has a unique white neighbor. Minimum temporal source sets based on time-respecting reachability [18] likewise do not depend on the colors of a source's other neighbors.

There is also a close structural boundary with temporal path covers. Every footprint-constrained process, after one parent is selected for each newly blue vertex, decomposes into vertex-disjoint strict temporal paths. Temporal path-cover problems optimize path collections subject to temporal reachability or disjointness [5, 6]; an arbitrary temporal path cover need not satisfy the footprint-uniqueness condition that makes a path a valid forcing chain. The footprint-constrained parameter can thus be viewed as a uniqueness-constrained temporal path-cover number, not a reformulation of the unconstrained path-cover problems.

Temporal controllability is a mature adjacent literature. In particular, zero forcing has already been used for time-varying structural perturbations [15], and direct graph conditions and input-set optimization are known for temporal strong structural controllability [16]. We make no claim here that either temporal color process characterizes a previously studied matrix class.

The one-layer boundary is also known under different language. If W is the set left white initially, one synchronous layer succeeds exactly when every $w \in W$ has an external private neighbor. Writing $\mathrm{OIR}(G)$ for the maximum cardinality of an open irredundant set, this gives the exact identity

$$Z_{\mathrm{LL}}((V, E_1)) = |V| - \mathrm{OIR}(G_1).$$

Open irredundance and its upper parameter are studied in [9]. Similarly, integer programming for bounded-time static zero forcing predates the present temporal model [3]. Our complexity contribution therefore uses a multi-layer temporal sequence in which each edge appears exactly once.

3 Definitions

All graphs are finite, simple, and undirected. A temporal graph is an ordered tuple

$$\mathcal{G} = (V, E_1, \ldots, E_T),$$

where $G_t = (V, E_t)$ is the layer at time t. The lifetime is T, and the *footprint* is

$$H = \left(V, \bigcup_{t=1}^{T} E_t\right).$$

Colors persist. If $B \subseteq V$ is the blue set immediately before layer t, define

$$F_t^{\mathrm{LL}}(B) = \{v \notin B : \exists u \in B, uv \in E_t, N_{G_t}(u) \setminus B = \{v\}\},$$
$$F_t^{\mathrm{FC}}(B) = \{v \notin B : \exists u \in B, uv \in E_t, N_H(u) \setminus B = \{v\}\}.$$

For $X \in \{\mathrm{LL}, \mathrm{FC}\}$, the synchronous process from a seed set S is

$$B_0^X = S, \qquad B_t^X = B_{t-1}^X \cup F_t^X(B_{t-1}^X).$$

Eligibility is always evaluated using B_{t-1}^X; newly blue vertices cannot force until the next layer.

Definition 3.1. A set $S \subseteq V$ is an X-temporal zero-forcing set if $B_T^X = V$. The minimum cardinality of such a set is $Z_X(\mathcal{G})$.

If several sources can force the same target in one round, the target is colored once. When a parent relation is needed, we choose one valid source.

4 General structure and sharp growth bounds

Proposition 4.1 (Dominance). *For every temporal graph $\mathcal{G}$,*

$$Z_{\mathrm{LL}}(\mathcal{G}) \leq Z_{\mathrm{FC}}(\mathcal{G}).$$

Proof. Run both processes from the same seed set. We prove inductively that $B_t^{\mathrm{FC}} \subseteq B_t^{\mathrm{LL}}$. Suppose u footprint-forces v at time t. Every footprint neighbor of u other than v is already blue in the footprint process and hence, by induction, in the layer-local process. If v is not already layer-locally blue, it is the unique white neighbor of u in G_t and is forced there as well. Thus every footprint-constrained forcing set is layer-local. □

Theorem 4.2 (Growth dichotomy)**.** *If $|V| = n$ and $\mathcal{G}$ has lifetime T, then*

$$Z_{\mathrm{LL}}(\mathcal{G}) \geq \left\lceil \frac{n}{2^T} \right\rceil$$

and

$$Z_{\mathrm{FC}}(\mathcal{G}) \geq \max\left\{ Z(H), \left\lceil \frac{n}{T+1} \right\rceil \right\}.$$

Both lifetime-dependent bounds are attained for infinitely many orders.

Proof. In a layer-local round each blue vertex has at most one unique white neighbor, so at most $|B_{t-1}^{\mathrm{LL}}|$ vertices are added. Hence $|B_t^{\mathrm{LL}}| \leq 2|B_{t-1}^{\mathrm{LL}}|$ and $n \leq 2^T|S|$.

In the footprint-constrained process, a vertex can force at most once over the entire lifetime. Immediately after it forces its unique footprint-white neighbor, every one of its footprint neighbors is blue permanently. After choosing one parent for every newly blue vertex, the forces therefore form vertex-disjoint directed paths rooted at the seeds. Time labels increase strictly along each path, so a path contains at most $T + 1$ vertices. Thus $n \leq (T + 1)|S|$. Moreover, every footprint-constrained force is a valid ordinary zero-forcing move in H, so S is a static zero-forcing set and $|S| \geq Z(H)$.

The layer-local bound is attained by the binomial temporal trees in Section 6. The footprint bound is attained by disjoint $(T+1)$-vertex paths whose successive edges appear at times $1, \ldots, T$. □

5 Exact alternating paths and cycles

Let P_n have vertices $1, \ldots, n$ in order. Define the two-layer temporal path $\mathcal{P}_n$ by

$$E_1 = \{(1, 2), (3, 4), \ldots\}, \qquad E_2 = \{(2, 3), (4, 5), \ldots\}.$$

Theorem 5.1 (Alternating paths)**.** *For every $n \geq 1$,*

$$Z_{\mathrm{LL}}(\mathcal{P}_n) = \left\lfloor \frac{n}{4} \right\rfloor + 1, \qquad Z_{\mathrm{FC}}(\mathcal{P}_n) = \left\lceil \frac{n}{3} \right\rceil.$$

Proof. For the layer-local rule, let $Q_k = \{2k - 1, 2k\} \cap [n]$. A seed in Q_k makes its first-layer pair blue and can then expand across both adjacent second-layer edges. Its complete reach is

$$I_k = \{2k - 2, 2k - 1, 2k, 2k + 1\} \cap [n].$$

Because both layers are matchings, multiple seeds color exactly the union of their intervals I_k. Each interval contains at most four vertices, so $\lceil n/4 \rceil$ seeds are necessary. When $n = 4a$, vertex 1 belongs only to the three-vertex interval I_1; consequently a intervals cover at most $3 + 4(a - 1) = n - 1$ vertices. This handles $n \equiv 0 \pmod 4$; combined with $\lceil n/4 \rceil$ for the other residues, it gives $\lfloor n/4 \rfloor + 1$ in every case. For the upper bound, write $n = 4a + r$. If $r = 0$, use the interval indices

$\{1, 3, \ldots, 2a-1, 2a\}$; if $r \in \{1, 2, 3\}$, use $\{1, 3, \ldots, 2a+1\}$. (For $a = 0$, the latter set is $\{1\}$.) These intervals cover $[n]$ with exactly $\lfloor n/4 \rfloor + 1$ choices; seed any vertex of the corresponding Q_k.

For the footprint-constrained rule, Theorem 4.2 with $T = 2$ gives the lower bound $\lceil n/3 \rceil$; the cases $n \leq 2$ are direct. For the upper bound, partition the first $3q$ vertices into triples $D_j = \{3j-2, 3j-1, 3j\}$. If j is odd, seed $3j-2$; if j is even, seed $3j$. For an internal odd block, the left neighbor outside the block is the root of the preceding even block and is already blue; hence the odd root has its middle vertex as unique white footprint neighbor and forces rightward in layers 1 and 2. Symmetrically, an internal even root is supported by the root of the following odd block and forces leftward. Endpoint blocks need no outside support. Thus q seeds color the first $3q$ vertices when $n = 3q$. If $n = 3q+1$, seed n; this also supports a final even block. If $n = 3q+2$ and q is even, seed $n-1$: it supports the final even root and forces n in layer 1. If q is odd, seed n, which forces $n-1$ in layer 2. These constructions use exactly $\lceil n/3 \rceil$ seeds. □

Since the footprint of $\mathcal{P}_n$ is P_n and $Z(P_n) = 1$, both temporal parameters exceed their aggregate static counterpart by a linear factor. The two temporal semantics themselves differ by $n/12 + O(1)$.

For even $n \geq 4$, define $\mathcal{C}_n$ by placing alternating perfect matchings of C_n in its two layers.

Theorem 5.2 (Alternating even cycles)**.** *For even $n \geq 4$,*

$$Z_{\mathrm{LL}}(\mathcal{C}_n) = \left\lceil \frac{n}{4} \right\rceil, \qquad Z_{\mathrm{FC}}(\mathcal{C}_n) = 2\left\lceil \frac{n}{6} \right\rceil.$$

Proof. The case $n = 4$ follows directly, so assume $n \geq 6$. For layer-local forcing, each seed in a first-layer pair reaches a cyclic arc of four vertices. Hence at least $\lceil n/4 \rceil$ seeds are necessary. Selecting alternating first-layer pairs covers the cycle with exactly that many arcs.

For the footprint-constrained upper bound, regard the $n/2$ second-layer edges as cyclic blocks. Seed both endpoints of every block in a minimum dominating set of this block cycle. In round one, the endpoints of a selected block force their outside first-layer neighbors. In round two, those newly blue vertices force the as-yet-uncolored endpoints of the two adjacent second-layer blocks. Thus a selected block covers itself and its two neighboring blocks, using $2\lceil (n/2)/3 \rceil = 2\lceil n/6 \rceil$ seeds.

The chain bound gives $n \leq 3|S|$. It remains only to exclude the odd value $|S| = 2q+1$ when $n = 6q+2$. Let f be the number of distinct vertices added in round one. Since E_1 is a matching, f is also the number of seeds that force in that round. If $f \leq |S| - 2$, then before round two there are $|S| + f$ blue vertices, but the f first-round sources can never force again under the footprint-constrained rule. Hence at most $|S|$ vertices can force in round two, and $n \leq |S| + f + |S| \leq 3|S| - 2$. Equality $f = |S|$ would require every seed to have exactly one blue cycle neighbor, so the seed-induced subgraph would be a disjoint union of edges and $|S|$ would be even. If $f = |S| - 1$, let x be the unique first-round nonforcer. If x has no blue cycle neighbor, its E_1 neighbor cannot become blue in round one and its only possible unique-white edge is unavailable in round two. If x has two blue neighbors, it has no white neighbor. If it has one blue neighbor, then every seed has degree one in the seed-induced subgraph, which would be 1-regular and would force $|S|$ to be even. Hence x cannot force in round two, so the second round adds at most $|S| - 1$ vertices and again $n \leq 3|S| - 2$. Thus $n = 6q+2$ requires at least $2q+2$ seeds, completing all residue classes. □

6 Aggregation, semantics, and time reversal

We next give one family witnessing three distinct failures of a static-only view. Start with $V_0 = \{r\}$. At time t, create one fresh child $u^{(t)}$ for every $u \in V_{t-1}$, put

$$L_t = \{u^{(t)} : u \in V_{t-1}\}, \quad E_t = \{uu^{(t)} : u \in V_{t-1}\}, \quad V_t = V_{t-1} \cup L_t.$$

Let $\mathcal{B}_T$ be the resulting temporal graph and H_T its footprint. Each layer is a matching, H_T is a tree, and $|V_T| = 2^T = n$.

Theorem 6.1 (Binomial temporal tree)**.** *For $T \geq 2$,*

$$Z_{\mathrm{LL}}(\mathcal{B}_T) = 1, \qquad Z_{\mathrm{FC}}(\mathcal{B}_T) = \frac{n}{2}, \qquad Z(H_T) = \frac{n}{4}.$$

If the layers are reversed, then

$$Z_{\mathrm{LL}}(\mathrm{rev}(\mathcal{B}_T)) = \frac{n}{2}.$$

Proof. Seed r. Inductively, all of V_{t-1} is blue before layer t, and every blue vertex has exactly one current white neighbor, its new child. Thus $Z_{\mathrm{LL}}(\mathcal{B}_T) = 1$.

In the first layer of the reversed process, E_T is a perfect matching between V_{T-1} and the leaves L_T. If an edge has neither endpoint seeded, its leaf can never be colored because the edge never reappears. Therefore every reversed layer-local forcing set has at least $|E_T| = n/2$ vertices. Seeding all leaves L_T colors their parents in the first reversed round and hence colors the whole graph. The same leaf set is valid footprint-constrained, while Proposition 4.1 supplies the lower bound. For the footprint-constrained forward lower bound, every forcing chain contains at most one vertex of L_T: the sole incident edge of such a leaf has label T, while labels increase strictly along a chain. Since $|L_T| = n/2$, at least $n/2$ chains and hence seeds are required. Directly, seeding V_{T-1} lets every old vertex force its time-T child, proving equality.

The leaves of H_T are exactly L_T, so there are $n/2$ of them. Static forcing chains form vertex-disjoint paths, each containing at most two tree leaves, and therefore $Z(H_T) \geq n/4$. For the reverse inequality, seed the time-T child of every vertex in V_{T-2}. These leaves force V_{T-2}; those vertices then force their time-$(T-1)$ children; finally the time-$(T-1)$ children force their own time-T children. The seed set has $|V_{T-2}| = n/4$ vertices. □

Together with Theorem 5.1, the family shows that static aggregation can either underestimate or overestimate layer-local temporal zero forcing by $\Theta(n)$.

An even simpler family gives the largest possible order of overestimation up to an additive constant.

Proposition 6.2 (Serialized stars)**.** *Let $\mathcal{S}_n$ be the temporal star with center c, leaves $v_1, \ldots, v_{n-1}$, and one-edge layers $E_t = \{cv_t\}$ in this order. For $n \geq 3$,*

$$Z_{\mathrm{LL}}(\mathcal{S}_n) = 1, \qquad Z_{\mathrm{FC}}(\mathcal{S}_n) = Z(K_{1,n-1}) = n - 2.$$

Proof. For the layer-local rule, seed the center; it forces one leaf in every layer. The footprint-constrained number is at least the static number $Z(K_{1,n-1}) = n-2$: at most one leaf can be omitted from a static forcing set, because two white leaves leave the center unable to distinguish either; seeding all but one leaf is sufficient. For temporal equality, seed $v_1, \ldots, v_{n-2}$. Vertex v_1 forces c in the first layer, and at the last layer c has only v_{n-1} white and forces it. □

Corollary 6.3 (Maximum additive aggregation overestimate)**.** *For every $n \geq 3$ there is a temporal graph $\mathcal{K}_n$ with*

$$Z_{\mathrm{LL}}(\mathcal{K}_n) = 1, \qquad Z_{\mathrm{FC}}(\mathcal{K}_n) = Z(K_n) = n - 1.$$

Consequently, the additive overestimate $Z(H) - Z_{\mathrm{LL}}(\mathcal{G})$ can attain its maximum possible value $n-2$.

Proof. First serialize the $n-1$ edges from a center c to the other vertices, and then place every remaining edge of K_n in its own later layer. Seeding c colors every other vertex during the initial star layers, so the later layers are irrelevant to layer-local forcing. The footprint is K_n, whose static zero-forcing number is $n-1$. The footprint lower bound gives $Z_{\mathrm{FC}} \geq n-1$. For the reverse inequality, seed every vertex except the last star leaf. At its star layer, c has that leaf as its unique white footprint neighbor and the corresponding edge is active, so c forces it. Finally, if a footprint has an edge then $Z(H) \leq n-1$ and $Z_{\mathrm{LL}} \geq 1$; for an edgeless footprint both numbers equal n. □

Static zero-forcing chains can be reversed to obtain a zero-forcing set of the same size [2]. The next result shows that this operation survives the temporal schedule under footprint-constrained forcing.

Theorem 6.4 (Footprint-constrained reversal)**.** *For every temporal graph,*

$$Z_{\mathrm{FC}}(\mathcal{G}) = Z_{\mathrm{FC}}(\mathrm{rev}(\mathcal{G})).$$

Proof. Take a successful forward process and choose one parent force for every newly blue vertex, discarding duplicate forces to the same target. By footprint uniqueness, every vertex has at most one selected outgoing force; every nonseed has one selected incoming force. The selected arcs are therefore vertex-disjoint directed paths or directed cycles. Their time labels increase strictly along each directed walk because a newly colored vertex cannot force in the same round. In particular, there is no directed cycle, so all components are paths. Every path has exactly one original seed and one terminal vertex, so the number of terminals equals the number of seeds.

Seed the reversed process with the terminal vertices of these paths and reverse every arc $x \to y$ used at time t to $y \to x$ at time $T+1-t$. We verify this schedule by induction over decreasing forward time labels. Vertex y is already blue then, either because it is terminal or because its later forward outgoing arc has already been reversed. Let $w \in N_H(y)\setminus\{x\}$. If w has no selected outgoing force, it is a terminal vertex and hence a seed of the reversed process. Otherwise let $w \to z$ occur at forward time s. If $s < t$, then y was white before time s; footprint uniqueness forces $z = y$, so y would become blue at time s, contradicting the selected force $x \to y$ at time t. If $s = t$, the source w can have only the unique target y; after selecting $x \to y$ as the parent of y, the duplicate $w \to y$ is not selected, contradicting the assumption that w has a selected outgoing force. Hence $s > t$, so $z \to w$ has already occurred in the prescribed reverse schedule. Thus every footprint neighbor of y other than x is blue. At reverse time $T+1-t$, either x has already become blue through an additional deterministic force, or x is the unique white footprint neighbor of y and the active edge xy forces it. By blue-set monotonicity, additional forces can only help; all prescribed targets are therefore blue by their scheduled reverse times. This proves $Z_{\mathrm{FC}}(\mathrm{rev}(\mathcal{G})) \leq Z_{\mathrm{FC}}(\mathcal{G})$; reversing twice gives equality. □

Theorem 6.1 shows that no analogous result holds for the layer-local rule: its reversal ratio can be $n/2$.

7 Computational complexity and parameterized algorithms

The decision problem LL-TZF asks whether $Z_{\mathrm{LL}}(\mathcal{G}) \leq K$. Membership in NP follows by simulating the deterministic process.

Theorem 7.1 (Restricted hardness)**.** LL-TZF *is NP-complete even when every layer has one nontrivial component that is a star, every footprint edge appears in exactly one layer, and the* *footprint is bipartite. Parameterized by K, it is W[2]-hard. The reduction satisfies*

$$\mathrm{OPT}_{\text{LL-TZF}} = \mathrm{OPT}_{\textsc{Set Cover}} + 1.$$

Proof. Let a Set Cover instance have universe $U = \{e_1, \ldots, e_p\}$ and sets $\mathcal{F} = \{F_1, \ldots, F_m\}$, with every element occurring in a set. Create vertices

$$\{c, a\} \cup \{s_1, \ldots, s_m\} \cup \{x_1, \ldots, x_p\}.$$

The first layer is the edge ca. For each element e_j, the next layer is the star centered at x_j with leaves s_i for which $e_j \in F_i$. Finally, for $i = 1, \ldots, m$, add a one-edge cleanup layer cs_i. Every edge appears once, and the footprint is bipartite with parts $\{c, x_1, \ldots, x_p\}$ and $\{a, s_1, \ldots, s_m\}$.

If C is a cover, seed c and the set vertices corresponding to C. Vertex c first forces a. At each element layer, a seeded incident set vertex forces the center x_j. The cleanup layers let c force all remaining set vertices. This gives a temporal forcing set of size $|C| + 1$.

Conversely, let S be feasible. At least one of c, a is seeded, since a has no active edge after the first layer. Let P index the set vertices that are blue before cleanup. Such a set vertex is either seeded or was forced in an element layer by its center. A center performing that force must itself be seeded, because it has no earlier incident edge. A center forced during its own element layer cannot force back in that layer under the synchronous update convention; moreover, any seeded center can force at most one set vertex. Therefore the vertices indexed by P can be injectively charged to seeds outside $\{c, a\}$.

If element e_j is not contained in any set indexed by P, then no incident set vertex was blue at its layer. Hence x_j must be seeded. It was not used in the preceding charge, since forcing an incident set vertex would put a set containing e_j into P. For every such uncovered element, choose one arbitrary containing set. Together with the sets indexed by P, these choices form a cover of size at most $|S| - 1$. Thus the optima differ by exactly one. The transformation sends k to $k + 1$; parameterized Set Cover is W[2]-complete [7], establishing W[2]-hardness. ☐

Using the classical NP-complete Vertex Cover problem [10] as the incidence instance makes every element star a P_3, yielding NP-completeness even when every layer has at most two edges.

Theorem 7.2 (Seed-budget–lifetime algorithms)**.** *Both temporal zero-forcing problems are fixed-parameter tractable in $k + T$. More precisely, after a growth-bound rejection, exhaustive seed enumeration runs in*

$$2^{O(k(T+1))} \operatorname{poly}(|I|)$$

for layer-local forcing and

$$2^{O(k \log(T+1))} \operatorname{poly}(|I|)$$

for footprint-constrained forcing. The resulting vertex kernels have $k2^T$ and $k(T + 1)$ vertices, respectively; under explicit layer encoding, the second is a polynomial kernel in $k + T$.

Proof. For a nonempty instance, $k = 0$ is immediately a no instance. Now suppose $k \geq 1$. First observe that the final blue set is monotone in the seed set. Indeed, if $B \subseteq B'$, any force available from B either has its target already in B' or remains available because enlarging the blue set can only remove white neighbors. Induction over the layers proves monotonicity for both semantics. Thus a solution of size at most $k < n$ can be extended to one of size exactly k; if $k \geq n$, the instance is trivially feasible.

By Theorem 4.2, reject a layer-local instance if $n > k2^T$. Otherwise enumerate the k-subsets. The standard binomial bound $\binom{n}{k} \leq (en/k)^k$ gives $2^{O(k(T+1))}$ candidates. For the footprint-constrained problem reject if $n > k(T+1)$; the same estimate gives $(e(T+1))^k = 2^{O(k\log(T+1))}$ candidates. Each candidate is checked by direct simulation in polynomial time. If the size test passes, retaining the instance gives the stated vertex bounds; otherwise output a fixed no instance. With an explicit temporal edge list, $T[k(T+1)]^2$ bounds the encoding size of the footprint-constrained kernel by a polynomial in $k + T$. $\square$

8 Computational methods and verification

We implemented the two synchronous updates directly from their definitions, without encoding the closed forms, and enumerated seed sets in increasing cardinality. The implementation also computes ordinary static zero forcing by repeated closure. The checked instances comprise

- alternating paths through $n = 20$;
- alternating even cycles through $n = 22$;
- forward and reversed binomial temporal trees through $T = 4$ ($n = 16$);
- serialized temporal stars through $n = 12$; and
- completed serialized cliques through $n = 8$.

All 127 recorded parameter values matched their corresponding predicted values. The code records an optimum witness for every instance and writes machine-readable CSV and JSON results. It additionally exhausts all 4,096 two-layer temporal graphs on four labeled vertices to check dominance and footprint-constrained reversal invariance (S1 File). These computations are verification rather than evidence used in place of proof.

9 Discussion

The semantic choice determines whether time-varying neighborhoods create branching or merely schedule static deductions. Layer-local forcing can exploit temporarily absent edges, so it may be dramatically easier than forcing on the aggregate graph. The same deadline can also make it dramatically harder, as the alternating paths demonstrate. Footprint-constrained forcing is more conservative: it always begins with a static zero-forcing set, has no branching, and is time-reversal invariant.

These distinctions caution against assigning a single “temporal zero forcing number” without specifying (i) the neighborhood in which uniqueness is evaluated, (ii) whether layers permit one round or closure, and (iii) whether simultaneous eligibility uses the pre-round coloring. Our results use one synchronous round per layer. Because the exact path and cycle layers are matchings, allowing closure within those individual layers would not change their values, but it changes the general model.

A next structural problem is an explicit dynamic program on temporally labeled paths and trees beyond the alternating families. A second direction is to identify a precisely stated time-varying matrix class for which one of the parameters provides a sound controllability certificate. Existing temporal controllability results mean that such a connection must be proved, not inferred from the static interpretation.

10 Conclusion

Zero forcing on a temporal graph has at least two natural meanings. Evaluating uniqueness in the current layer allows repeated branching and strong temporal asymmetry; evaluating it in the footprint preserves static forcing chains and time-reversal symmetry. Exact families, aggregation gaps, complexity, and parameterized algorithms show that this distinction is mathematical rather than terminological. The results provide a starting point for further work on temporal graph classes and carefully delimited control applications.

Data and code availability

All code and machine-readable results underlying the computational verification are publicly available from the development repository on GitHub. The same verification materials are provided as S1 File. No external datasets were used.

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Supporting information

S1 File. Code and machine-readable verification results. ZIP archive containing the Python verification script, generated CSV and JSON results, reproduction instructions, tested Python version, and MIT license.